

Simulation

Section # 5

Problem 1

Write down the mean and variance for discrete and continuous distributions.

Answer

- Discrete distribution :-

Mean (Expected value) :- $E(X) = \sum_{i=0}^{\infty} X_i P(X_i)$

Variance :- $V(X) = E(X^2) - (E(X))^2$

1. Continuous distribution :-

Mean (Expected value) :- $E(X) = \int_{-\infty}^{\infty} xf(x)dx$

Variance :- $V(X) = \int_{-\infty}^{\infty} x^2 f(x)dx - (E(X))^2$

Problem 2

Determine mean and variance for tossing a single die, let X a random variable that counts the spots of the side facing up, it follows the following probability :

X	1	2	3	4	5	6
P(X=x)	1/21	2/21	3/21	4/21	5/21	6/21

Answer

Mean (Expected value):-

$$E(X) = (1)(1/21) + (2)(2/21) + (3)(3/21) + (4)(4/21) + (5)(5/21) + (6)(6/21) = 91/21$$

Variance V(X):-

$$E(X^2) = (1)^2(1/21) + (2)^2(2/21) + (3)^2(3/21) + (4)^2(4/21) + (5)^2(5/21) + (6)^2(6/21) = 21$$

Then $V(X) = 21 - (91/21)^2 = 2.222$

Problem 3

Drive the CDF of a distribution whose p.d.f is given by

$$f(x) = 1/200(x-20) \quad 20 \leq x \leq 40$$

$$0 \quad \text{otherwise}$$

Also get 3 R.Os following the above distribution using the following R.N (0.49 , 0.64 , 0.25)

Answer

a-To get CDF :

$$F(x) = \int_a^x f(x) dx$$

$$F(x) = \int_{20}^x \frac{1}{200} (x - 20) dx = \frac{1}{200} \int_{20}^x (x - 20) dx$$

$$= \frac{1}{200} \cdot \frac{(x-20)^2}{2} \Big|_{20}^x$$

F(x) when x=x

$$F(x=x) = \frac{(x-20)^2}{400}$$

F(X) when x=20

$$F(x = 20) = 0$$

$$F(x) = F(x) - F(20) = \frac{(x-20)^2}{400}$$

$$F(x) = \frac{(x-20)^2}{400}$$

To get random observation(R.O) $\longrightarrow$ $R=F(X)$

$$R = \frac{(x-20)^2}{400} \rightarrow 400 R = (x - 20)^2 \rightarrow 20\sqrt{R} = x - 20$$

$x = 20\sqrt{R} + 20$

When x = 20

$$20 = 20\sqrt{R} + 20$$

$$\sqrt{R} = 0 \rightarrow R = 0$$

When x = 40

$$40 = 20\sqrt{R} + 20 \rightarrow R = 1 \quad \text{Then } 0 \leq R \leq 1$$

To get R.O :-

$$R.N : 0.49 \rightarrow x_1 = 20\sqrt{0.49} + 20 = 34, \quad 0 \leq R \leq 1 \quad 20 \leq x \leq 40$$

$$R.N : 0.64 \rightarrow x_2 = 20\sqrt{0.64} + 20 = 36, \quad 0 \leq R \leq 1 \quad 20 \leq x \leq 40$$

$$R.N : 0.25 \rightarrow x_3 = 20\sqrt{0.25} + 20 = 30, \quad 0 \leq R \leq 1 \quad 20 \leq x \leq 40$$

→ Note : X must be in that range $20 \leq x \leq 40$